

Building and Testing Series and Parallel Circuits

PhET Circuit Construction Kit: DC · PhET

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/circuit-construction-kit-dc>

Curriculum links: circuits · electricity

Learning intentions

- I can construct series and parallel circuits using batteries, resistors and bulbs.
- I can use an ammeter and voltmeter to measure current and voltage at different points in a circuit.
- I can explain how current and voltage behave differently in series versus parallel circuits.

Getting started

Open PhET Circuit Construction Kit: DC. Drag a battery, at least two resistors or light bulbs, and connecting wires into the workspace to build a simple series circuit. Add the ammeter and voltmeter from the toolbox so you can take readings as you go.

Tasks

1. Build a simple series circuit with one battery and one resistor. Record the current reading from the ammeter and the voltage reading across the battery.

2. Add a second resistor in series (in the same loop) with the first. Record the new current reading. Has it increased, decreased or stayed the same compared to Task 1?

3. Build four different series circuits by varying the number or resistance of resistors. Record the current and voltage for each.

Circuit setup	Current (A)	Voltage across battery (V)

4. Using your table, describe the relationship between total resistance and current in a series circuit, assuming voltage stays constant.

5. Rebuild your circuit so the two resistors are in parallel rather than series. Measure the current through each individual branch, then measure the current leaving the battery.

6. Add the two branch currents from Task 5 together. Compare this total to the current leaving the battery. What circuit law does this demonstrate?

Working:

7. Measure the voltage across each resistor in your parallel circuit. Compare this to the voltage across each resistor in your original series circuit from Tasks 1-2. Explain the difference.

8. A string of old Christmas lights wired in series all go out when one bulb fails, but modern parallel-wired lights don't. Using what you've built and measured in this simulation, explain why.

Extension (optional)

Design a circuit in the simulation with one battery, two resistors in parallel, and that parallel combination in series with a third resistor. Predict the current through each resistor before building it, then check your prediction using the meters.
